

GURU NANAK COLLEGE (AUTONOMOUS)

Chennai – 600 042.



BACHELOR OF COMPUTER SCIENCE COMPUTER SCIENCE PROGRAMME

2024 – 2025

Name : _____

Reg. No : _____

Class : _____

Year : **III YEAR** **Semester: V SEMESTER**

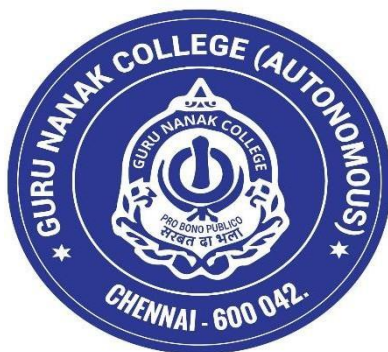
Subject Code : **20UCSC312P**

Subject Name : **PYTHON PROGRAMMING LAB**

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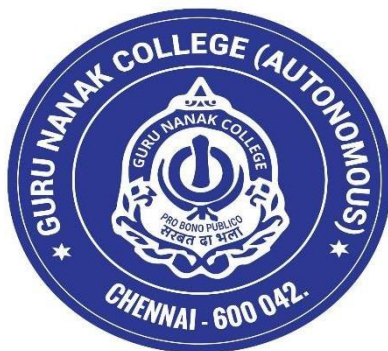
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COMPUTER SCIENCE PROGRAMME

BONAFIDE CERTIFICATE

NAME : _____
REG. NO : _____
CLASS : _____

This is to certify that this is the bonafide record of the practical work done in _____ at Guru Nanak College Computer Lab, during the Year 2024 - 2025

Staff-In-Charge

Head of the Department

*Submitted for the _____
B.Sc., Computer Science Practical Examination held on _____
at Guru Nanak College, Chennai – 42.*

Internal Examiner

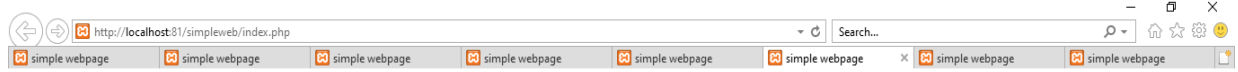
External Examiner

S.NO	DATE	CONTENTS	PAGE NO	SIGN
1		Creating simple webpage using PHP		
2		Use of conditional Statements in PHP		
3		Use of Looping Statements in PHP		
4		Creating different types of arrays		
5		File manipulations using PHP		
6		Creations of Sessions		
7		Creations of Cookies		
8		Simple Applications using PHP with Input Validations		
9		Create Simple Table with Constraints		
10		Insertion, Updating and Deletion of row in MySQL		
11		Searching of Data by Different criteria		
12		Sorting of Data		
13		Demonstration of Joining tables		
14		Usage of aggregate functions		
15		Database connectivity in PHP with MySQL		

SIMPLE WEBPAGE USING PHP

```
<html>
<title> simple webpage </title>
<body style="background_colour:OFFFF;color:red;">
    <?php
        $website=array("creating website using HTML","Creating website using PHP","creating
website using wordpress");
        $item=rand(0,sizeof($website)-1);
        ?>
        <h1><?php
            echo $website[$item]
                ?>
        </h1>
    </body>
</html>
```

Output Screen:



creating website using wordpress



USE OF CONDITIONAL STATEMENT

If...Else If...Else Statement

```
<html>

<head>

<title></title>

</head>

<body>

<?php

// defining a variable

$marks=75;

if($marks>79)

{

echo"GRADE :A";

}

elseif($marks>=79&& $marks>60)

{

echo"GRADE :B";

}

elseif($marks>=60&& $marks>50){

echo"GRADE :C";

}elseif($marks=50){

echo"GRADE :D";

}else{

echo"GRADE :F";

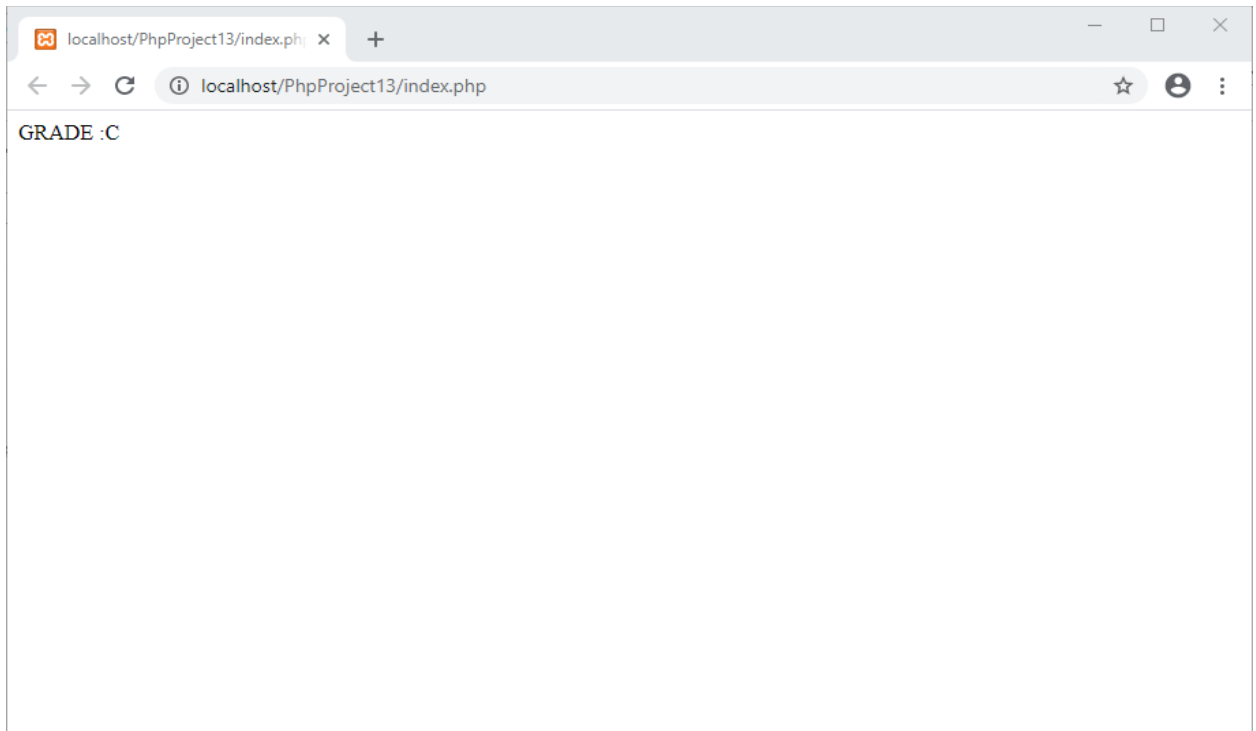
}

?>

</body>

</html>
```

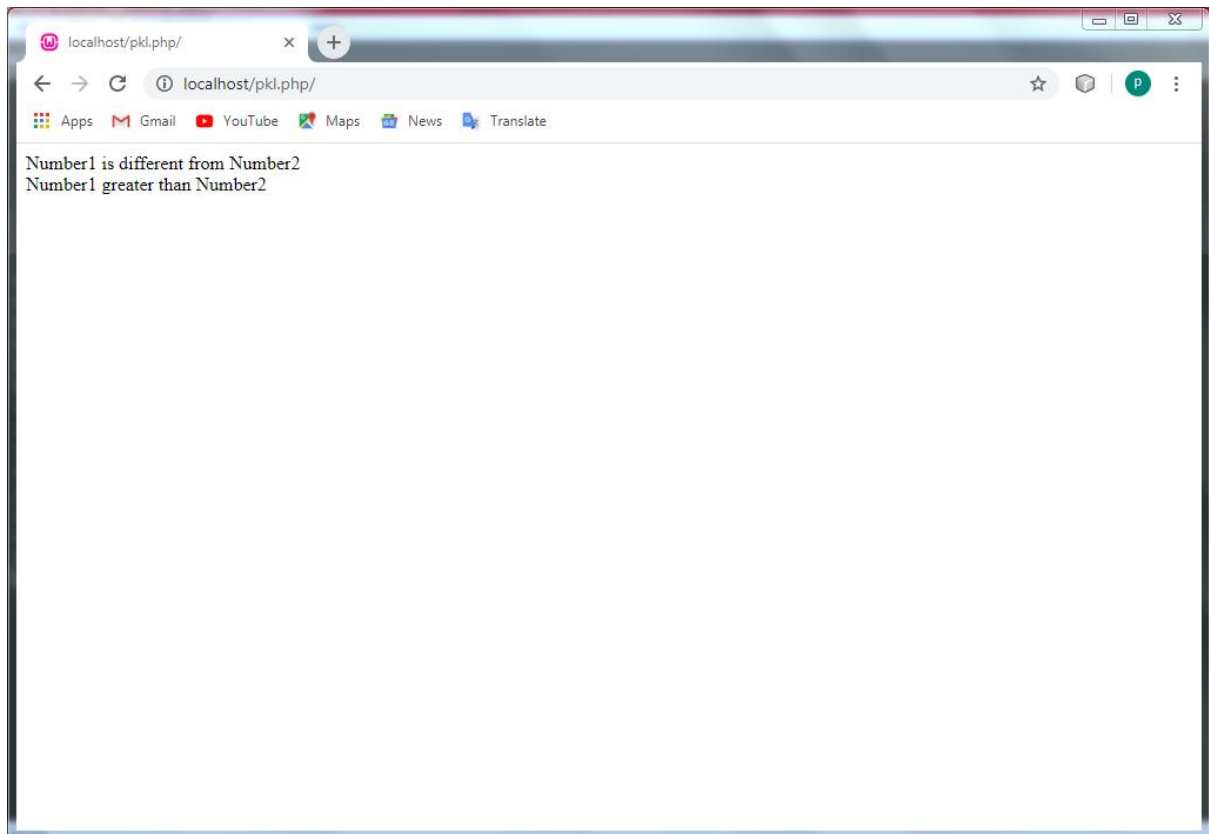
Output Screen:



Nested If Else

```
<html>
<head>
<title></title>
</head>
<body>
<?php
$number1=40;
$number2=12;
if($number1!=$number2)
{
echo"Number1 is different from Number2";
echo "<br>";
if($number1>$number2)
{
echo"Number1 greater than Number2";
}
else
{
echo"Number2 is grater than Number1";
}
}
else
{
echo"Number1 is equal to Number2";
}
?>
</body>
</html>
```

Output Screen:



Switch Case Statement

```
<html>

<head>

<title></title>

</head>

<body>

<?php

$gender = "M";

switch ($gender) {

case "F":

echo "F is Female";

break;

case "M":

echo "M is Male";

break;

default:

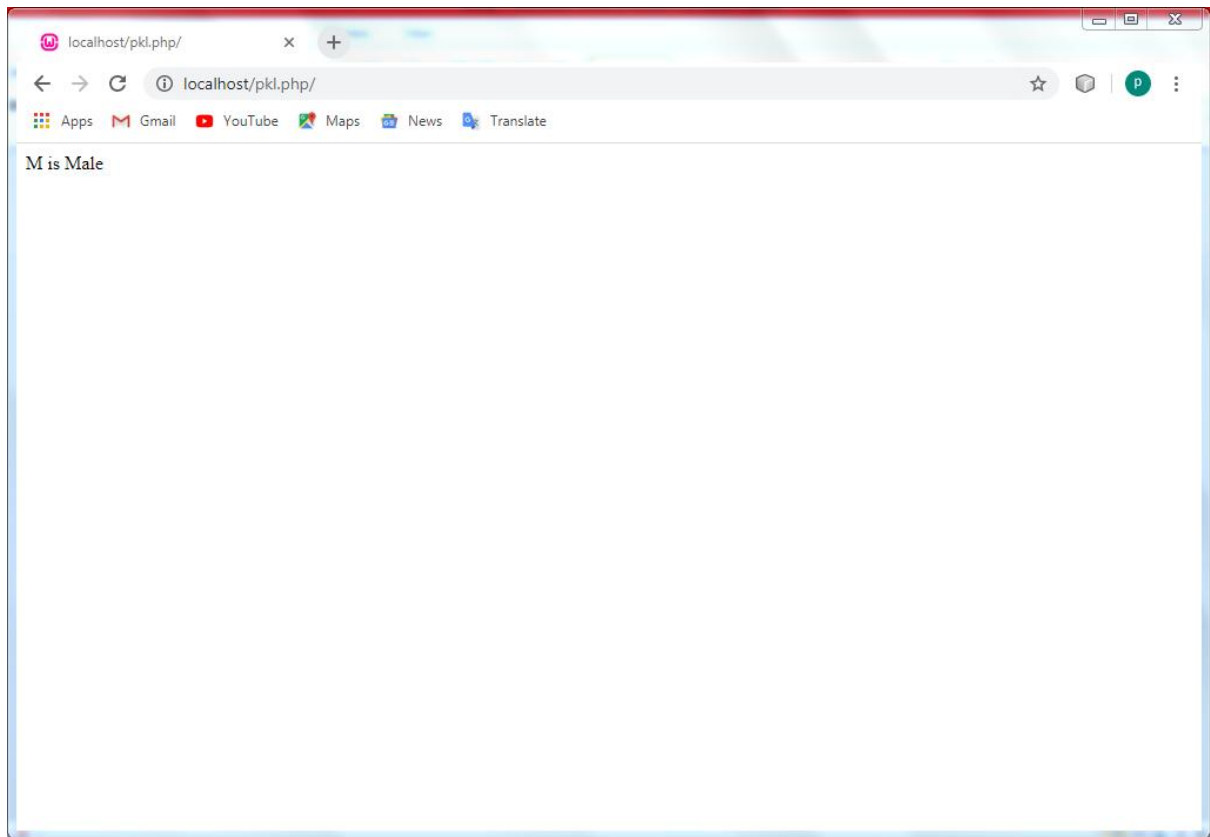
echo "invalid";

}

?>

</body>

</html>
```

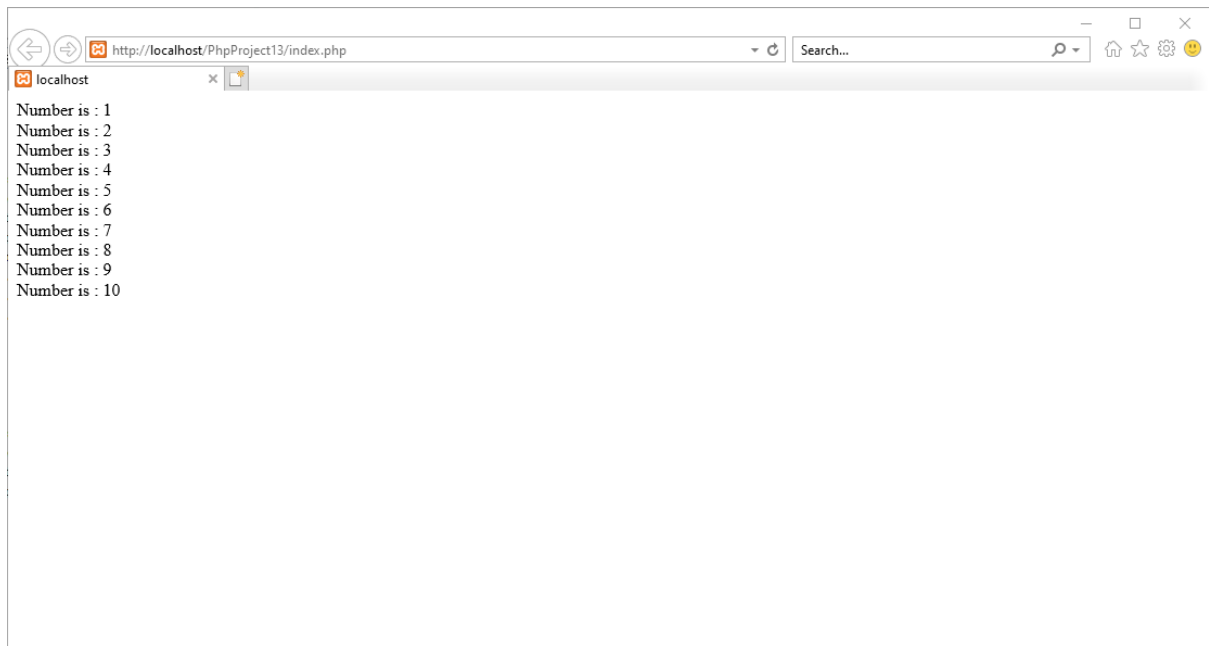


USE OF LOOPING STATEMENTS

While:

```
<html>
  <head>
    <title>While</title>
  </head>
  <body>
    <?php
      $x=1;
      while($x<=10)
      {
        echo"number is :$x<br>";
        $x++;
      }
    ?>
  </body>
</html>
```

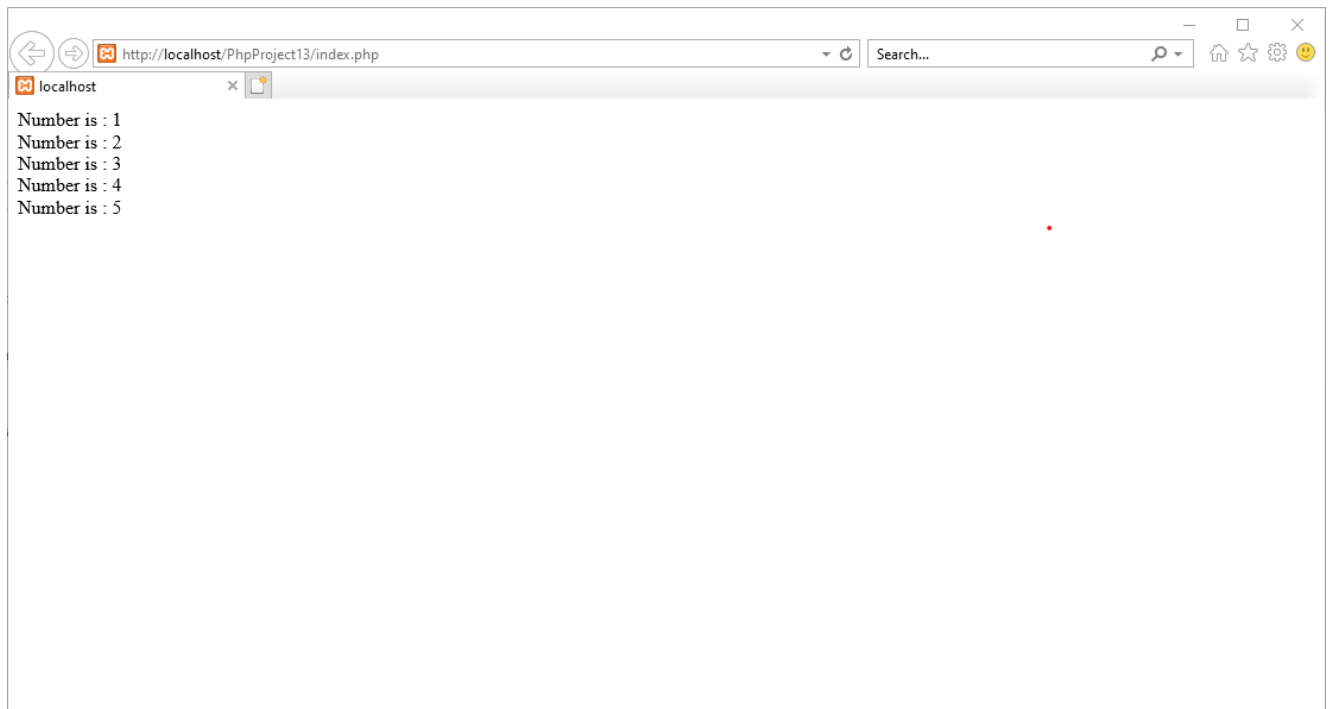
Output Screen:



Do.. while...

```
<html>
  <head>
    <title>Do while</title>
  </head>
  <body>
    <?php
      $x=1;
      do
      {
        echo"Number is : $x <br>";
        $x++;
      }
      while($x<=5)
    ?>
  </body>
</html>
```

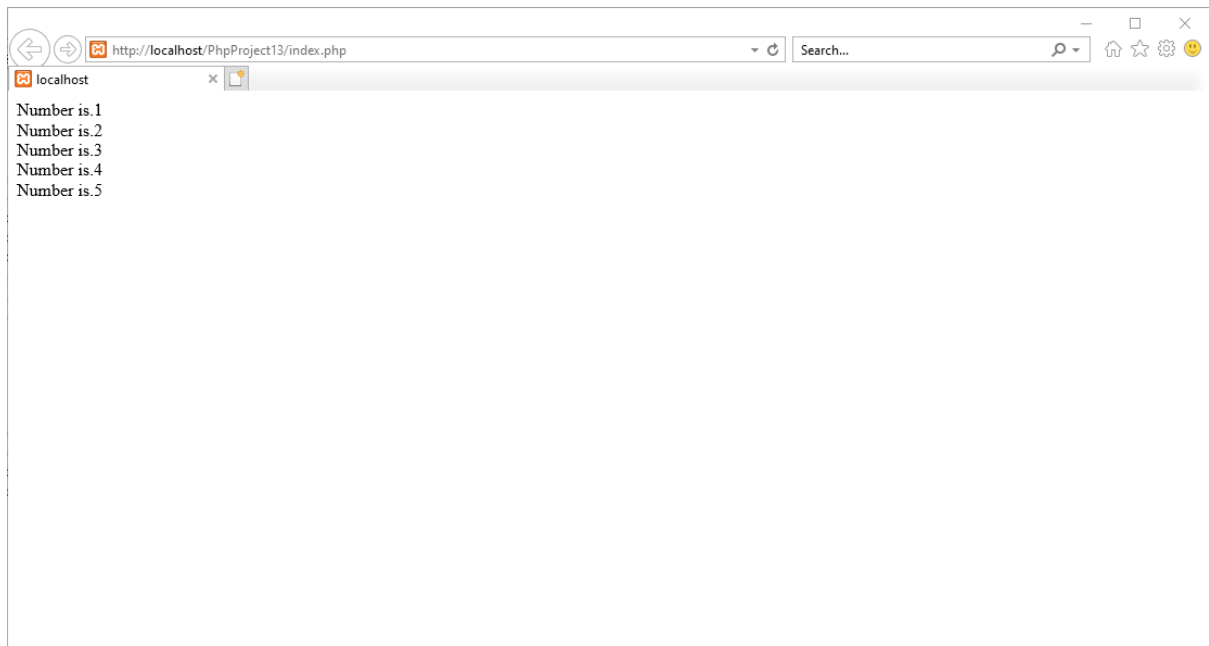
Output Screen:



For... loop...

```
<html>
<head>
<title>for loop</title>
</head>
<body>
<?php
for($x=1;$x<=5;$x++)
{
echo"Number is.$x<br>";
}
?>
</body>
</html>
```

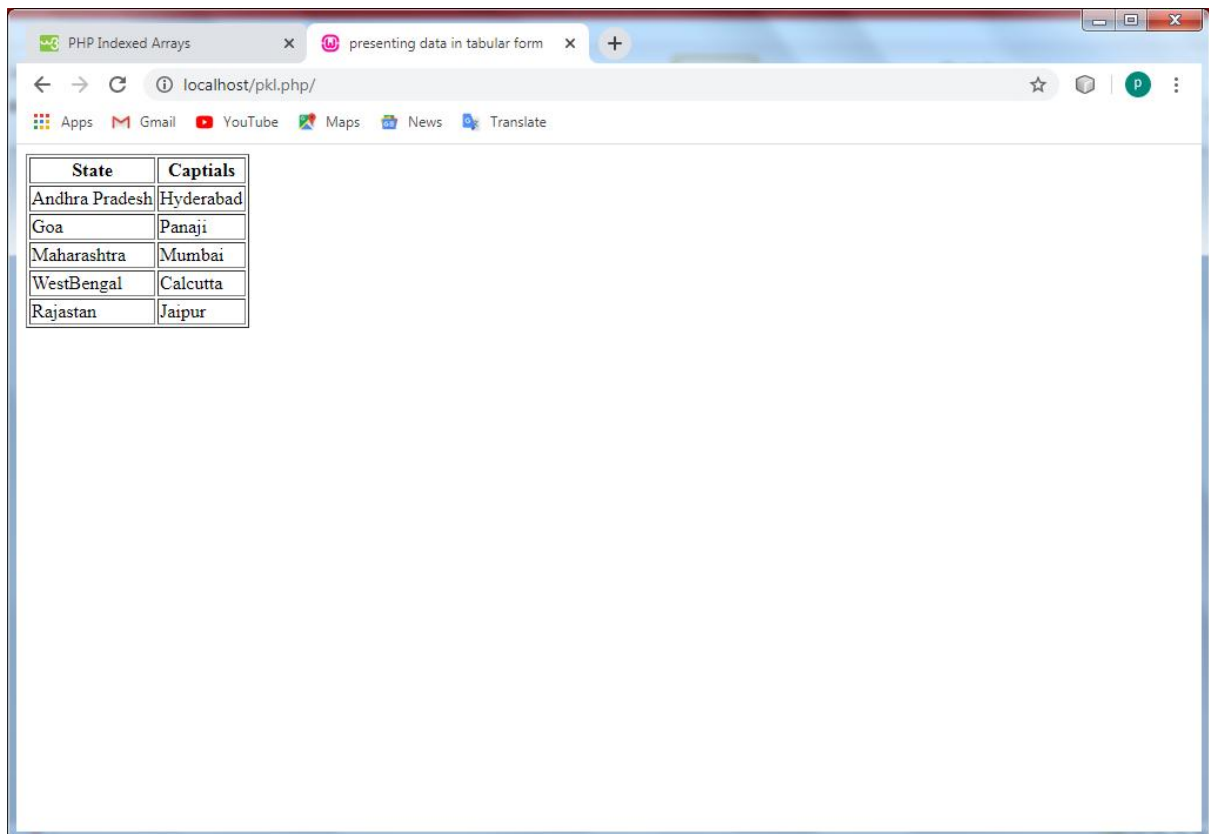
Output Screen:



ARRAY PROGRAM

```
<html>
<head>
<title>_Creating a Array Program
</title>
</head>
<body>
<html>
<head>
<title>presenting data in tabular form</title>
</head>
<body>
<?php
$array=array('Andhra
Pradesh'=>'Hyderabad','Goa'=>'Panaji','Maharashtra'=>'Mumbai','WestBengal'=>'Calcutta','R
ajasthan'=>'Jaipur');
echo"<table border= '1'>";
echo"<tr>";
echo"<th>State</th>";
echo"<th>Captials</th>";
echo"</tr>";
reset($array);
while(list($key,$value)=each($array))
{
echo"<tr>";
echo"<td>$key</td>";
echo"<td>$value</td>";
echo"</tr>";
}
echo"</table>";
?>
</body>
</html>
```

Output Screen:



The screenshot shows a web browser window with two tabs. The first tab is titled 'PHP Indexed Arrays' and the second is 'presenting data in tabular form'. The address bar shows 'localhost/pkl.php/'. Below the browser interface, a table is displayed with two columns: 'State' and 'Captials'. The table contains five rows of data representing Indian states and their capitals.

State	Captials
Andhra Pradesh	Hyderabad
Goa	Panaji
Maharashtra	Mumbai
WestBengal	Calcutta
Rajasthan	Jaipur

FILE MANIPULATION USING PHP

Index.php

```
<html>

<head>

    <meta charset="UTF-8">

    <title></title>

</head>

<body>

    <?php

        echo readfile("webdictionary.txt");

        echo("<br><br>");


        $myfile = fopen("webdictionary.txt", "r") or die("Unable to open file!");

        // Output one line until end-of-file

        while(!feof($myfile)) {

            echo fgets($myfile) . "<br>";

        }

        $myfile = fopen("webdictionary.txt", 'w');

        $text = "Hello world\n";

        fwrite($myfile, $text);

        fclose($myfile);

    ?>

</body>

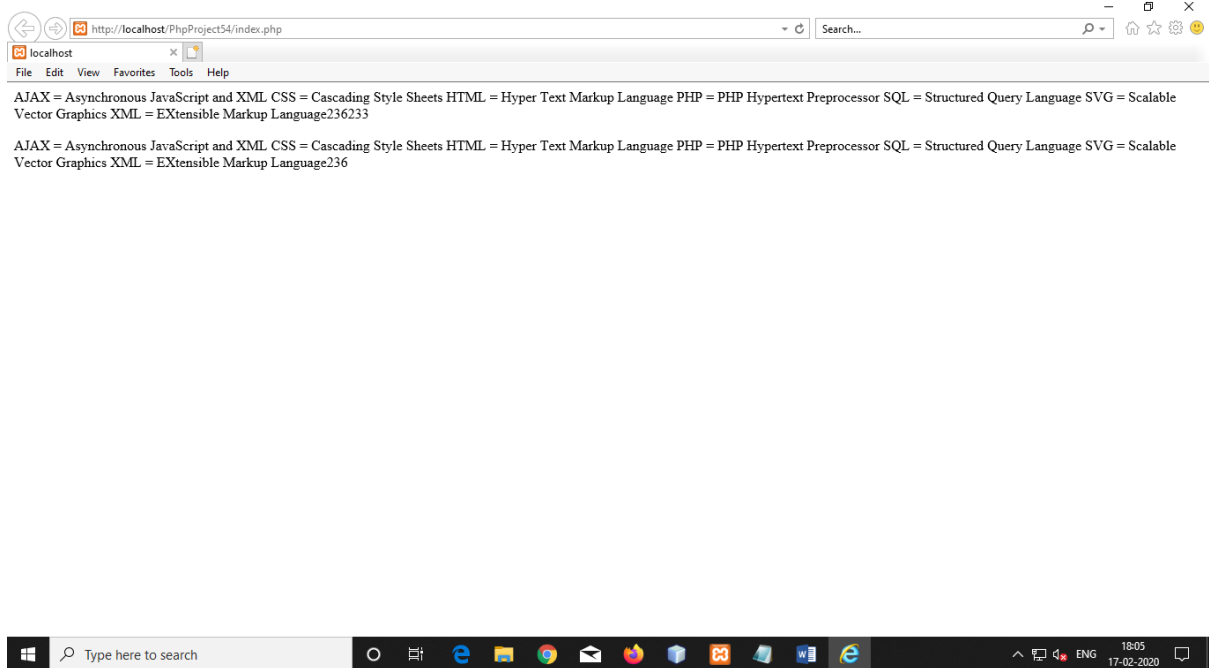
</html>
```

Webdictionary.txt

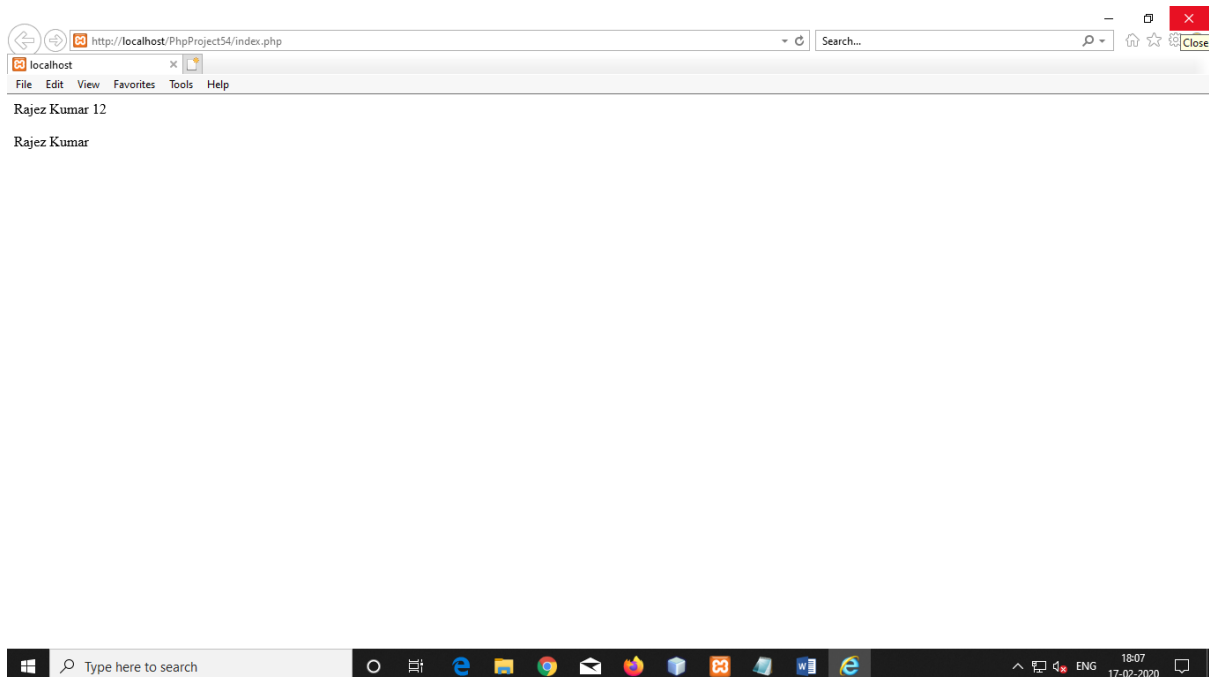
AJAX = Asynchronous JavaScript and XML CSS = Cascading Style Sheets HTML = Hyper Text Markup Language PHP = PHP Hypertext Preprocessor SQL = Structured Query Language SVG = Scalable Vector Graphics XML = EXtensible Markup Language

Output Screen:

Reading File



Writing File:



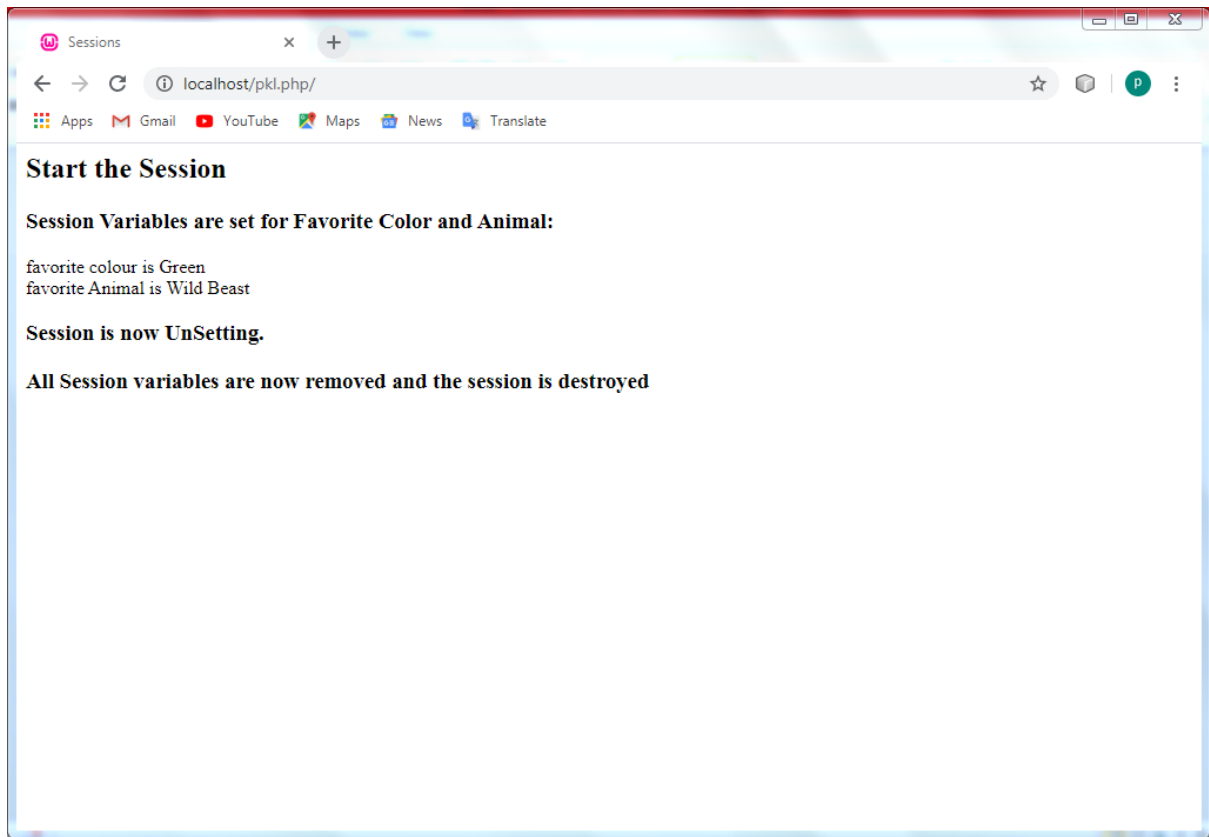
PHP PROGRAM TO CREATE AND DESTROY SESSIONS

```
<html>
<head>
<title></title>
</head>
<body>
<?php
    session_start();
?>
<html>
<head>
<meta charset="UTF-8">
<title>Sessions</title>
</head>
<body>
<?php
    //set session variables
    echo "<h2><b>Start the Session<br></b><h2>";
    $_SESSION["favcolor"]="Green";
    $_SESSION["favanimal"]="Wild Beast";
    echo "<h3><b>Session Variables are set for Favorite Color and Animal:<br></b></h3>";
    echo "favorite colour is ". $_SESSION["favcolor"]."<br>";
    echo "favorite Animal is ". $_SESSION["favanimal"]."<br>";
    //remove all session variable;
    session_unset();
    echo "<h3><b>Session is now UnSetting.<br></b></h3>";
    //destroy the session
    session_destroy();

    echo "<h3><b>All Session variables are now removed and the session is
    destroyed<br></b></h3>";
?>

</body>
</html>
```

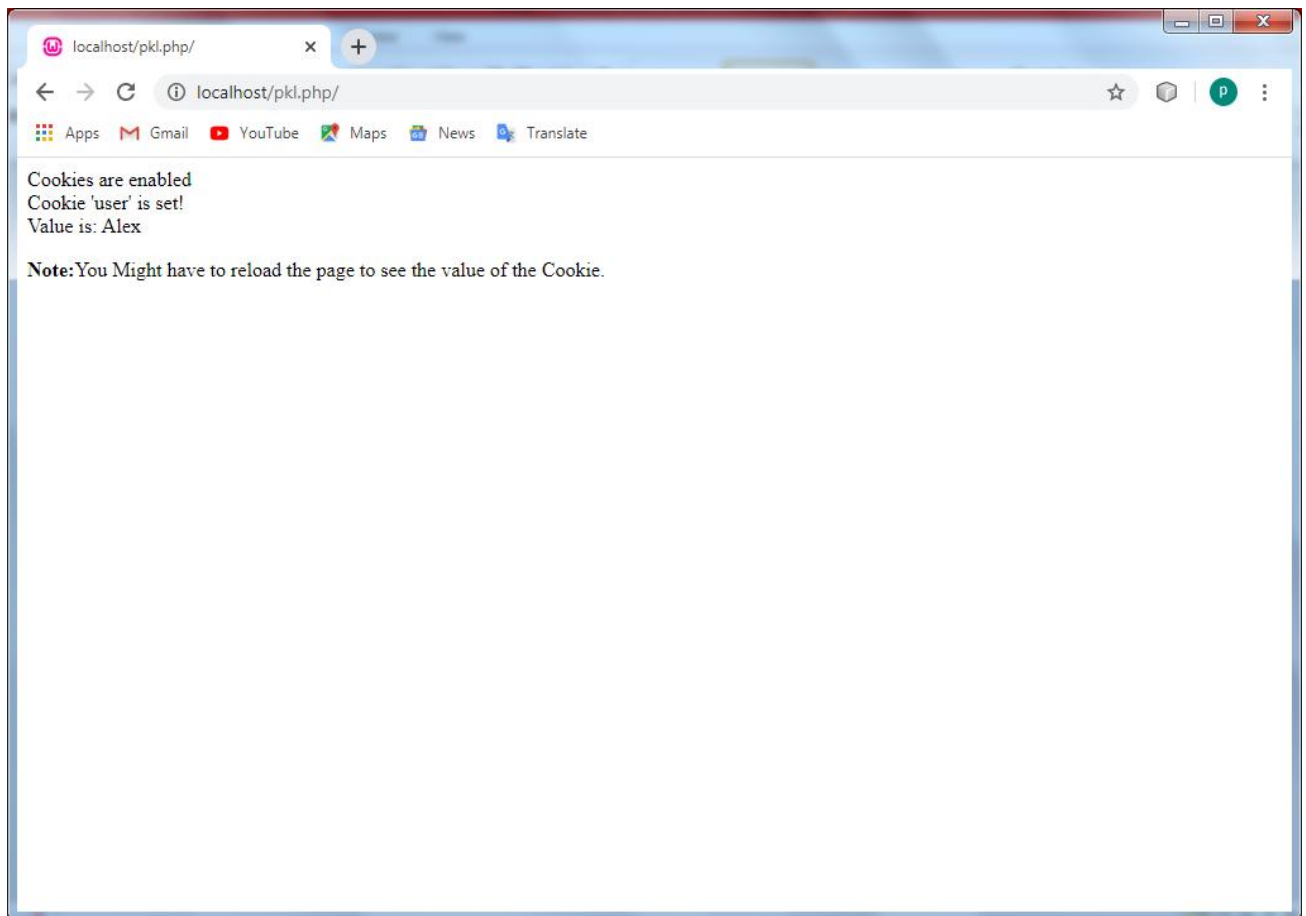
Output Screen:



CREATION OF COOKIES

```
<html>
<head>
<title>cookies</title>
</head>
<body>
<?php
$cookie_name='user';
$cookie_value="Alex";
setcookie("$cookie_name","$cookie_value",time()+(86400*30),'/'); //86400 = 1 Day
?>
<html>
<body>
<?php
if(count($_COOKIE)>0){
echo"Cookies are enabled<br>";
}else{
echo"Cookies are disabled<br>";}
if(!isset($_COOKIE[$cookie_name])){
echo "Cookie Named '". $cookie_name.'" is not set!";
}else{
echo"Cookie '". $cookie_name.'" is set!<br>";
echo"Value is: ". $_COOKIE[$cookie_name];
}
?>
<p><strong>Note:</strong>You Might have to reload the page to see the value of the
Cookie.</p>
</body>
</html>
```

Output Screen:



CREATING SIMPLE APPLICATION USING PHP WITH INPUT VALIDATION

```
<html>

<head>

<title></title>

</head>

<body>

<?php

// define variables and set to empty values

$name = $email = $gender = $comment = $website = "";

if ($_SERVER["REQUEST_METHOD"] == "POST") {

    $name = test_input($_POST["name"]);

    $email = test_input($_POST["email"]);

    $website = test_input($_POST["website"]);

    $comment = test_input($_POST["comment"]);

    $gender = test_input($_POST["gender"]);

}

function test_input($data) {

    $data = trim($data);

    $data = stripslashes($data);

    $data = htmlspecialchars($data);

    return $data;

}

?>

<h2>PHP Form Validation Example</h2>

<form method="post" action="<?php echo
htmlspecialchars($_SERVER["PHP_SELF"]);?>">

Name:<input type="text" name="name">

<br><br>

E-mail: <input type="text" name="email">

<br><br>
```

Website: <input type="text" name="website">

Comment: <textarea name="comment" rows="5" cols="40"></textarea>

Gender:

<input type="radio" name="gender" value="female">Female

<input type="radio" name="gender" value="male">Male

<input type="radio" name="gender" value="other">Other

<input type="submit" name="submit" value="Submit">

</form>

<?php

echo "<h2>Your Input:</h2>";

echo \$name;

echo "
";

echo \$email;

echo "
";

echo \$website;

echo "
";

echo \$comment;

echo "
";

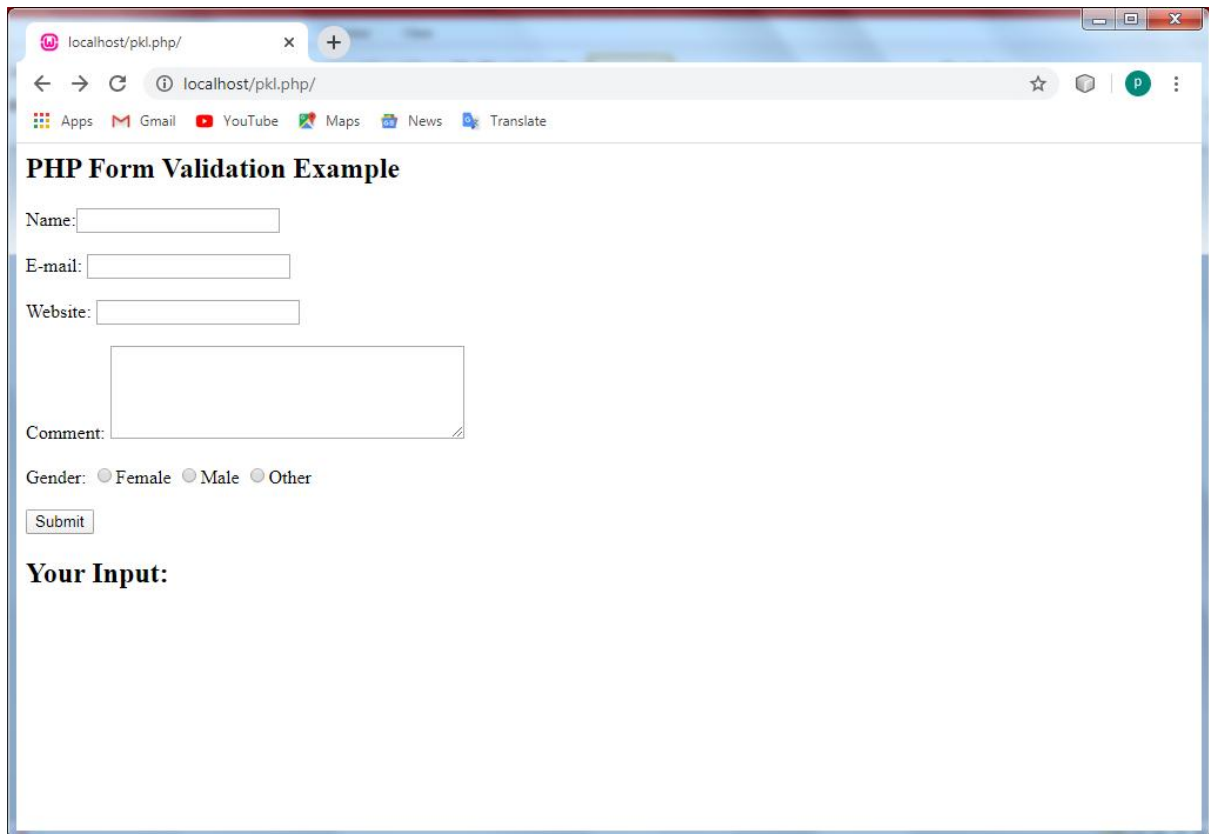
echo \$gender;

?>

</body>

</html>

Output Screen:



localhost/pkl.php/

localhost/pkl.php/

Apps Gmail YouTube Maps News Translate

PHP Form Validation Example

Name:

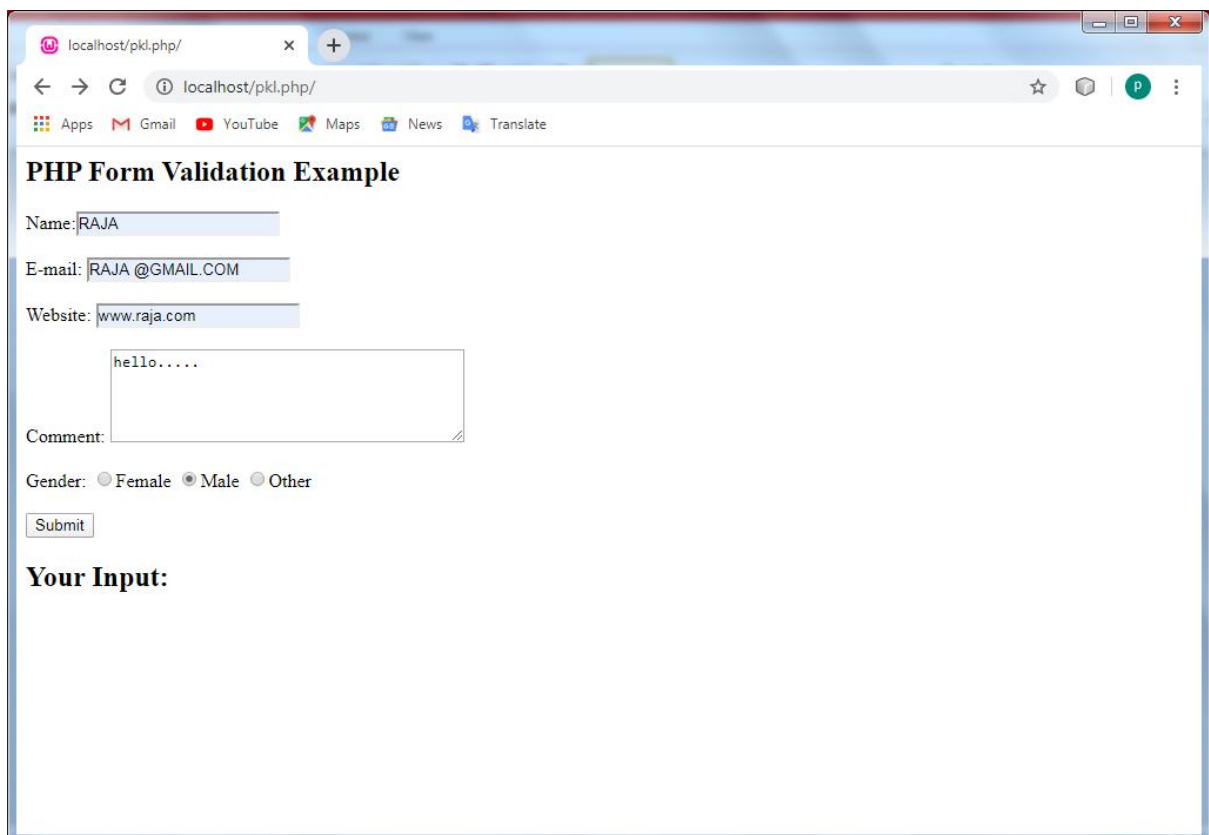
E-mail:

Website:

Comment:

Gender: ☐ Female ☐ Male ☐ Other

Your Input:



localhost/pkl.php/

localhost/pkl.php/

Apps Gmail YouTube Maps News Translate

PHP Form Validation Example

Name:

E-mail:

Website:

Comment:

Gender: ☐ Female ☒ Male ☐ Other

Your Input:

localhost/pkl.php/

localhost/pkl.php/

AppsGmailYouTubeMapsNewsTranslate

PHP Form Validation Example

Name:

E-mail:

Website:

Comment:

Gender: ☐ Female ☐ Male ☐ Other

Submit

Your Input:

RAJA

RAJA @GMAIL.COM

www.raja.com

hello.....

male

DATABASE CONNECTIVITY IN PHP WITH MYSQL

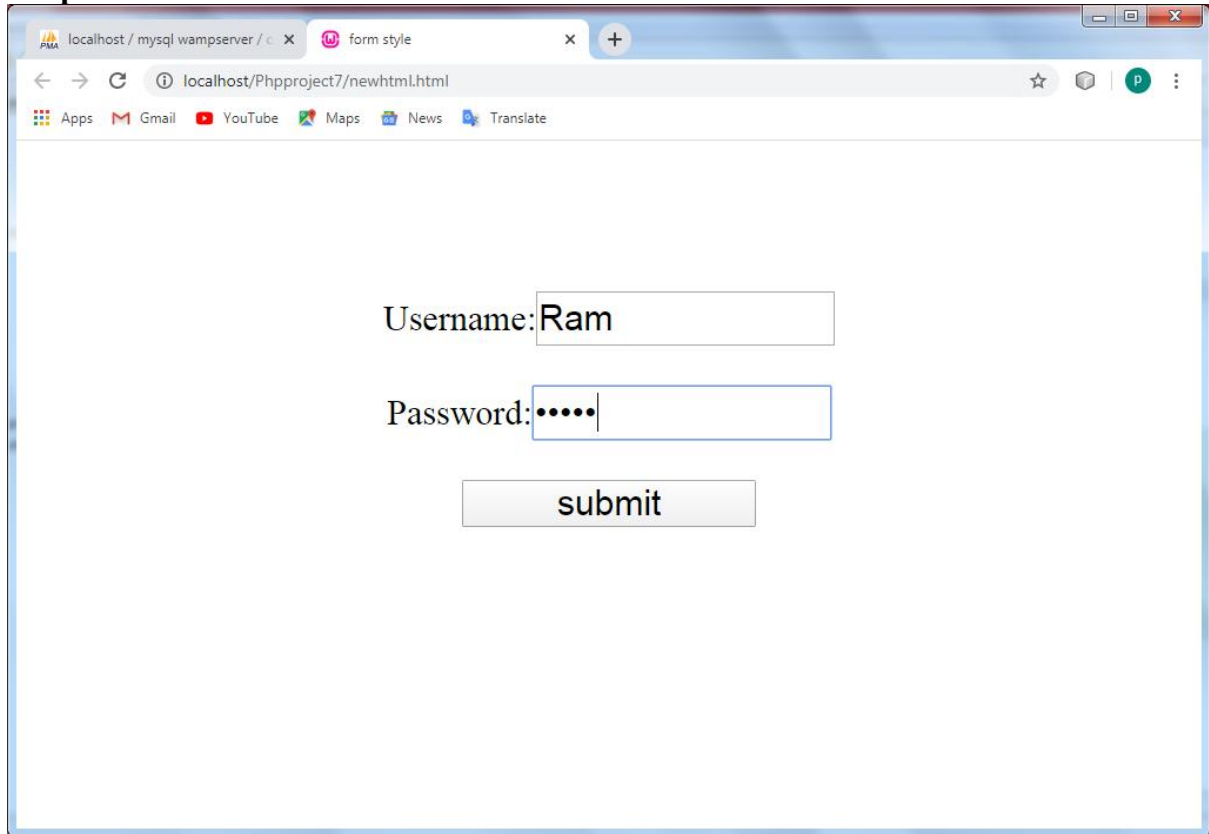
Form.html

```
<html>
<head>
<title>
form style</title>
<style>
form
{
padding-top:120px;
text-align:center;
font-size:30px;
}
input{
width:250px;
height:40px;
font-size:30px;
</style>
</head>
<body>
<form method="post" action="connect.php">
Username:<input type="text" name="username"><br><br>
Password:<input type="password" name=password><br><br>
<input type="submit" value="submit">
</form>
</body>
</html>
```

Connect.php

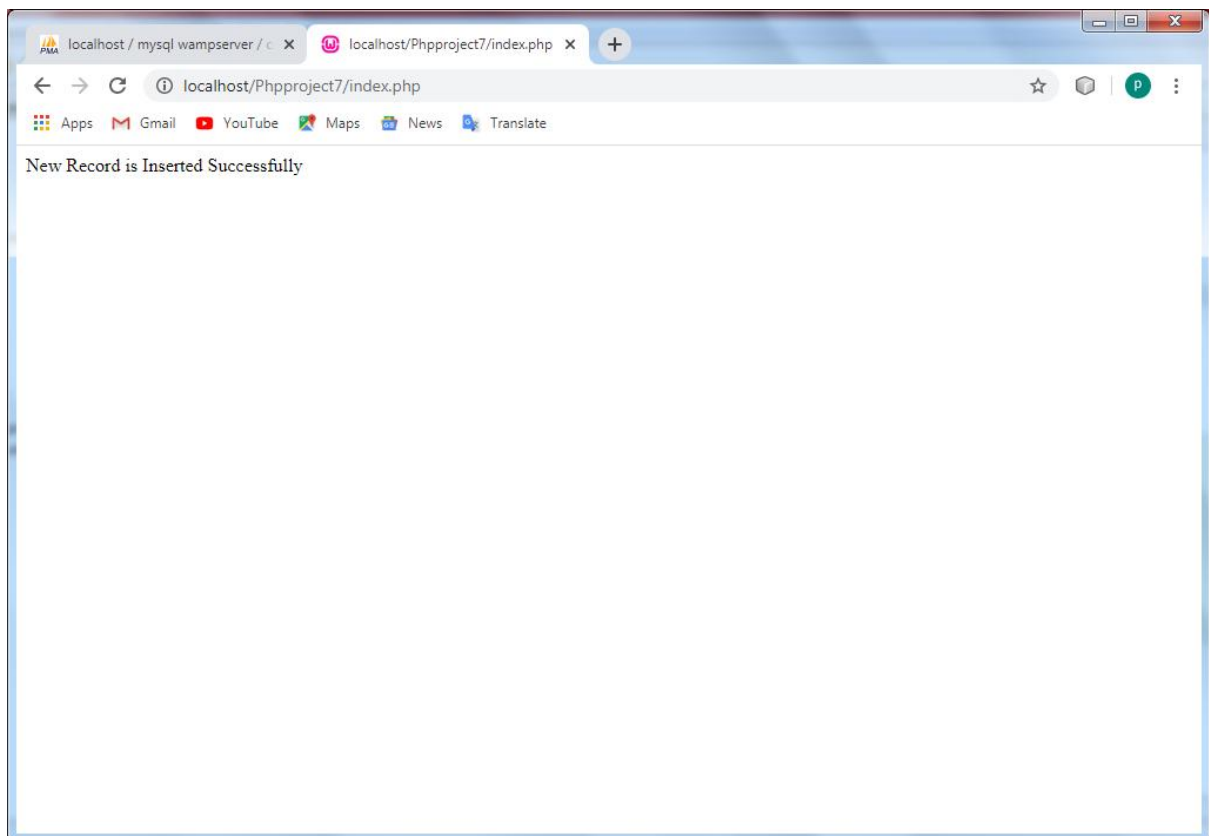
```
<html>
<head>
    <title>connection</title>
</head>
<body>
<?php
$username=filter_input(INPUT_POST,'username');
$password=filter_input(INPUT_POST,'password');
if(!empty($username)){
if(!empty($password)){
$host="localhost";
$dbusername="root";
$dbpassword="";
$dbname="connectdb";
$conn=new mysqli($host,$dbusername,$dbpassword,$dbname);
if(mysqli_connect_error()){
die('connect error(' .mysqli_connect_errno().')'.mysqli_connect_error());
}
else
{
$sql="INSERT INTO account(username,password) values('$username',$password)";
if($conn->query($sql)
{
echo"new rrecord is inserted successfully";
}
else
{
echo"error:".sql."<br>".$conn->error;
}
$conn->close();
}
}
else
{
echo"password should not be empty";
die();
}
}
else
{
echo"username should not be empty";
die();
}
?>
</body>
</html>
```

Output Screen:



A screenshot of a web browser window. The address bar shows 'localhost/Phpproject7/newhtml.html'. The page contains a login form with the following elements:

- A label 'Username:' followed by a text input field containing the text 'Ram'.
- A label 'Password:' followed by a password input field containing five dots '.....'.
- A 'submit' button located below the password field.



A screenshot of a web browser window. The address bar shows 'localhost/Phpproject7/index.php'. The page displays a confirmation message:

New Record is Inserted Successfully

localhost / mysql wampserver / c x localhost/Phpproject7/index.php x

localhost/phpmyadmin/sql.php?db=connectdb&table=account&server=1&target=&token=24893df6decff6fbe2...

Apps Gmail YouTube Maps News Translate

phpMyAdmin

(Recent tables) ...

- New
- connectdb
 - New
 - account
- information_schema
- mysql
- performance_schema
- rock
- test

Server: mysql wampserver » Database: connectdb » Table: account

Browse Structure SQL Search Insert Export Import More

⚠ Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

✔ Showing rows 0 - 0 (1 total, Query took 0.0000 sec)

```
SELECT * FROM `account`
```

☐ Profiling [Inline] [Edit] [Explain SQL] [Create PHP Code] [Refresh]

Number of rows: 25 ▾

+ Options

username	password
Ram	12345

Number of rows: 25 ▾

Query results operations

Print view Print view (with full texts) Export Display chart Create view

AGGREGATE FUNCTIONS

```
mysql> SELECT MIN(Roll_number) FROM student_mark;
```

```
+-----+
| MIN(Roll_number) |
+-----+
|          101 |
+-----+
```

```
1 row in set (0.05 sec)
```

```
mysql> SELECT MAX(Roll_number) FROM student_mark;
```

```
+-----+
| MAX(Roll_number) |
+-----+
|          105 |
+-----+
```

```
1 row in set (0.00 sec)
```

```
mysql> SELECT MIN(Roll_number) FROM student_mark where total<=200;
```

```
+-----+
| MIN(Roll_number) |
+-----+
|          105 |
+-----+
```

```
1 row in set (0.00 sec)
```

```
mysql> SELECT COUNT(Roll_number) FROM student_mark;
```

```
+-----+
| COUNT(Roll_number) |
+-----+
|           4 |
+-----+
```

```
1 row in set (0.03 sec)
```

```
mysql> SELECT COUNT(Roll_number) FROM student_mark WHERE total>200;
```

```
+-----+
| COUNT(Roll_number) |
+-----+
|           3 |
+-----+
```

```
1 row in set (0.00 sec)
```

```
mysql> SELECT AVG(total) FROM student_mark;
```

```
+-----+
| AVG(total) |
+-----+
|    271.5000 |
+-----+
```

AGGREGATE FUNCTIONS

+-----+

1 row in set (0.01 sec)

```
mysql> SELECT sum(total) FROM student_mark;
```

+-----+

| sum(total) |

+-----+

| 1086 |

+-----+

1 row in set (0.01 sec)

```
mysql>
```

TABLE CREATION

```
mysql> create table student_mark(student_name varchar(30), Roll_number int(20), mark1 int(20), mark2 int(20), mark3 int(20), mark4 int(20));
```

Query OK, 0 rows affected (2.48 sec)

```
mysql> desc student_mark;
```

```
mysql> desc student_mark;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| student_name | varchar(30)   | YES  |     | NULL    |       |
| Roll_number  | int(20)       | YES  |     | NULL    |       |
| mark1        | int(20)       | YES  |     | NULL    |       |
| mark2        | int(20)       | YES  |     | NULL    |       |
| mark3        | int(20)       | YES  |     | NULL    |       |
| mark4        | int(20)       | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.28 sec)

mysql>
```

6 rows in set (0.28 sec)

```
mysql> create table telephone(name varchar(20) not null, number bigint(20), country_code int(20), address varchar(20), primary key(number));
```

Query OK, 0 rows affected (0.91 sec)

```
mysql> desc telephone;
```

```
mysql> desc telephone;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| name       | varchar(20)   | NO   |     | NULL    |       |
| number     | bigint(20)    | NO   | PRI | NULL    |       |
| country_code | int(20)       | YES  |     | NULL    |       |
| address    | varchar(20)   | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.04 sec)

mysql> _
```

DELETION OF ROWS IN TABLE

```
mysql> delete from student_mark where Roll_number=102;
```

Query OK, 1 row affected (0.20 sec)

```
mysql> select * from student_mark;
```

```
mysql> select * from student_mark;
+-----+-----+-----+-----+-----+-----+-----+
| student_name | Roll_number | mark1 | mark2 | mark3 | mark4 | total |
+-----+-----+-----+-----+-----+-----+-----+
| Rajesh      | 101         | 70    | 64    | 53    | 90    | 277    |
| jagan       | 103         | 90    | 75    | 96    | 99    | 360    |
| Isai        | 104         | 56    | 45    | 74    | 80    | 255    |
| Saran       | 105         | 24    | 67    | 43    | 60    | 194    |
+-----+-----+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)

mysql>
```

GROUP BY Commands

```
mysql> SELECT student_name from student_mark where total>200 group by  
student_name order by student_name asc;
```

```
+-----+  
| student_name |
```

```
+-----+
```

```
| Isai      |
```

```
| jagan     |
```

```
| Rajesh    |
```

```
+-----+
```

```
3 rows in set (0.05 sec)
```

```
mysql> SELECT student_name from student_mark where total>200 group by  
student_name order by student_name desc;
```

```
+-----+  
| student_name |
```

```
+-----+
```

```
| Rajesh     |
```

```
| jagan      |
```

```
| Isai       |
```

```
+-----+
```

```
3 rows in set (0.00 sec)
```

```
mysql>
```

JOIN & INNER JOIN

mysql> select * from emp1;

```
+-----+-----+-----+
| eid | name  | sal  |
+-----+-----+-----+
| 101 | Rajesh | 25000 |
| 102 | Karthiga | 67000 |
| 103 | jagan  | 43000 |
| 104 | balaji | 150000 |
| 105 | soundu | 15000 |
+-----+-----+-----+
5 rows in set (0.00 sec)
```

mysql> select * from emp2;

```
+-----+-----+-----+
| eid | name  | sal  |
+-----+-----+-----+
| 201 | Kumar  | 5000 |
| 202 | Indu   | 15000 |
| 203 | nathan | 17000 |
| 203 | vijay  | 7000 |
| 205 | prakash | 17000 |
+-----+-----+-----+
5 rows in set (0.00 sec)
```

mysql> select a.eid,a.name,a.sal,b.eid,b.name,b.sal from emp1 a, emp2 b where a.sal=b.sal;

```
+-----+-----+-----+-----+-----+
| eid | name  | sal  | eid | name  | sal  |
+-----+-----+-----+-----+-----+
| 105 | soundu | 15000 | 202 | Indu   | 15000 |
+-----+-----+-----+-----+-----+
1 row in set (0.02 sec)
```

mysql> select a.eid,a.name,a.sal,b.eid,b.name,b.sal from emp1 a, emp2 b where a.sal>b.sal;

```
+-----+-----+-----+-----+-----+
| eid | name  | sal  | eid | name  | sal  |
+-----+-----+-----+-----+-----+
| 101 | Rajesh | 25000 | 201 | Kumar  | 5000 |
| 102 | Karthiga | 67000 | 201 | Kumar  | 5000 |
| 103 | jagan  | 43000 | 201 | Kumar  | 5000 |
| 104 | balaji | 150000 | 201 | Kumar  | 5000 |
+-----+-----+-----+-----+-----+
```

JOIN & INNER JOIN

```
| 105 | soundu | 15000 | 201 | Kumar | 5000 |
| 101 | Rajesh | 25000 | 202 | Indu | 15000 |
| 102 | Karthiga | 67000 | 202 | Indu | 15000 |
| 103 | jagan | 43000 | 202 | Indu | 15000 |
| 104 | balaji | 150000 | 202 | Indu | 15000 |
| 101 | Rajesh | 25000 | 203 | nathan | 17000 |
| 102 | Karthiga | 67000 | 203 | nathan | 17000 |
| 103 | jagan | 43000 | 203 | nathan | 17000 |
| 104 | balaji | 150000 | 203 | nathan | 17000 |
| 101 | Rajesh | 25000 | 203 | vijay | 7000 |
| 102 | Karthiga | 67000 | 203 | vijay | 7000 |
| 103 | jagan | 43000 | 203 | vijay | 7000 |
| 104 | balaji | 150000 | 203 | vijay | 7000 |
| 105 | soundu | 15000 | 203 | vijay | 7000 |
| 101 | Rajesh | 25000 | 205 | prakash | 17000 |
| 102 | Karthiga | 67000 | 205 | prakash | 17000 |
| 103 | jagan | 43000 | 205 | prakash | 17000 |
| 104 | balaji | 150000 | 205 | prakash | 17000 |
+-----+-----+-----+-----+-----+-----+
22 rows in set (0.00 sec)
```

```
mysql> select * from emp1 inner join emp2 on emp1.eid=emp2.eid;
Empty set (0.00 sec)
```

```
mysql> select * from emp1 inner join emp2 on emp1.sal=emp2.sal;
+-----+-----+-----+-----+-----+-----+
| eid | name | sal | eid | name | sal |
+-----+-----+-----+-----+-----+-----+
| 105 | soundu | 15000 | 202 | Indu | 15000 |
+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

```
mysql> select emp1.eid,emp2.name from emp1 inner join emp2 on emp1.sal=emp2.sal;
+-----+-----+
| eid | name |
+-----+-----+
| 105 | Indu |
+-----+-----+
1 row in set (0.00 sec)
```

INSERT TABLE

```
mysql> insert into student_mark(student_name, Roll_number, mark1, mark2, mark3, mark4) values("Rajesh", 101,70,64,53,90);
```

Query OK, 1 row affected (0.44 sec)

```
mysql> insert into student_mark(student_name, Roll_number, mark1, mark2, mark3, mark4) values("Karthiga", 102,90,85,84,100);
```

Query OK, 1 row affected (0.09 sec)

```
mysql> insert into student_mark(student_name, Roll_number, mark1, mark2, mark3, mark4) values("jagan", 103,90,75,96,99);
```

Query OK, 1 row affected (0.14 sec)

```
mysql> insert into student_mark(student_name, Roll_number, mark1, mark2, mark3, mark4) values("Isai", 104,56,45,74,80);
```

Query OK, 1 row affected (0.13 sec)

```
mysql> insert into student_mark(student_name, Roll_number, mark1, mark2, mark3, mark4) values("Saran", 105,24,67,43,60);
```

Query OK, 1 row affected (0.14 sec)

```
mysql> select * from student_mark;
```

student_name	Roll_number	mark1	mark2	mark3	mark4
Rajesh	101	70	64	53	90
Karthiga	102	90	85	84	100
jagan	103	90	75	96	99
Isai	104	56	45	74	80
Saran	105	24	67	43	60

5 rows in set (0.00 sec)

OUTER JOIN

```
mysql> select * from emp1;
```

```
+-----+-----+-----+
| eid | name  | sal  |
+-----+-----+-----+
| 101 | Rajesh | 25000 |
| 102 | Karthiga | 67000 |
| 103 | jagan  | 43000 |
| 104 | balaji | 150000 |
| 105 | soundu | 15000 |
+-----+-----+-----+
5 rows in set (0.00 sec)
```

```
mysql> select * from emp2;
```

```
+-----+-----+-----+
| eid | name  | sal  |
+-----+-----+-----+
| 201 | Kumar | 5000 |
| 202 | Indu  | 15000 |
| 203 | nathan | 17000 |
| 203 | vijay | 7000 |
| 205 | prakash | 17000 |
+-----+-----+-----+
5 rows in set (0.00 sec)
```

```
mysql> select * from emp1,emp2 where emp1.eid=emp2.eid ;
```

Empty set (0.01 sec)

```
mysql> select * from emp1,emp2 where emp1.sal>emp2.sal;
```

```
+-----+-----+-----+-----+-----+-----+
| eid | name  | sal  | eid | name  | sal  |
+-----+-----+-----+-----+-----+-----+
| 101 | Rajesh | 25000 | 201 | Kumar | 5000 |
| 102 | Karthiga | 67000 | 201 | Kumar | 5000 |
| 103 | jagan  | 43000 | 201 | Kumar | 5000 |
| 104 | balaji | 150000 | 201 | Kumar | 5000 |
```

OUTER JOIN

105 soundu 15000 201 Kumar 5000
101 Rajesh 25000 202 Indu 15000
102 Karthiga 67000 202 Indu 15000
103 jagan 43000 202 Indu 15000
104 balaji 150000 202 Indu 15000
101 Rajesh 25000 203 nathan 17000
102 Karthiga 67000 203 nathan 17000
103 jagan 43000 203 nathan 17000
104 balaji 150000 203 nathan 17000
101 Rajesh 25000 203 vijay 7000
102 Karthiga 67000 203 vijay 7000
103 jagan 43000 203 vijay 7000
104 balaji 150000 203 vijay 7000
105 soundu 15000 203 vijay 7000
101 Rajesh 25000 205 prakash 17000
102 Karthiga 67000 205 prakash 17000
103 jagan 43000 205 prakash 17000
104 balaji 150000 205 prakash 17000

+-----+-----+-----+-----+-----+-----+

22 rows in set (0.00 sec)

SEARCH OF DATA BY DIFFERENT CRITERIA

mysql> select * from student_mark;

```
+-----+-----+-----+-----+-----+-----+
| student_name | Roll_number | mark1 | mark2 | mark3 | mark4 | total |
+-----+-----+-----+-----+-----+-----+
| Rajesh      | 101 | 70 | 64 | 53 | 90 | 277 |
| jagan       | 103 | 90 | 75 | 96 | 99 | 360 |
| Isai        | 104 | 56 | 45 | 74 | 80 | 255 |
| Saran       | 105 | 24 | 67 | 43 | 60 | 194 |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```

mysql> select * from student_mark where mark1= 90;

```
+-----+-----+-----+-----+-----+-----+
| student_name | Roll_number | mark1 | mark2 | mark3 | mark4 | total |
+-----+-----+-----+-----+-----+-----+
| jagan        | 103 | 90 | 75 | 96 | 99 | 360 |
+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

mysql> select * from student_mark where mark1!=90;

```
+-----+-----+-----+-----+-----+-----+
| student_name | Roll_number | mark1 | mark2 | mark3 | mark4 | total |
+-----+-----+-----+-----+-----+-----+
| Rajesh      | 101 | 70 | 64 | 53 | 90 | 277 |
| Isai        | 104 | 56 | 45 | 74 | 80 | 255 |
| Saran       | 105 | 24 | 67 | 43 | 60 | 194 |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

mysql> select * from student_mark where mark1<90;

```
+-----+-----+-----+-----+-----+-----+
| student_name | Roll_number | mark1 | mark2 | mark3 | mark4 | total |
+-----+-----+-----+-----+-----+-----+
| Rajesh      | 101 | 70 | 64 | 53 | 90 | 277 |
| Isai        | 104 | 56 | 45 | 74 | 80 | 255 |
| Saran       | 105 | 24 | 67 | 43 | 60 | 194 |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

mysql> select * from student_mark where mark1 in(100);

Empty set (0.00 sec)

SEARCH OF DATA BY DIFFERENT CRITERIA

mysql> select * from student_mark where mark1 between 55 and 80;

```
+-----+-----+-----+-----+-----+-----+
| student_name | Roll_number | mark1 | mark2 | mark3 | mark4 | total |
+-----+-----+-----+-----+-----+-----+
| Rajesh      | 101         | 70    | 64    | 53    | 90    | 277    |
| Isai        | 104         | 56    | 45    | 74    | 80    | 255    |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

mysql> select * from student_mark where mark1=70 AND MARK2=64;

```
+-----+-----+-----+-----+-----+-----+
| student_name | Roll_number | mark1 | mark2 | mark3 | mark4 | total |
+-----+-----+-----+-----+-----+-----+
| Rajesh      | 101         | 70    | 64    | 53    | 90    | 277    |
+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

mysql> select * from student_mark where mark1=70 OR MARK2=64;

```
+-----+-----+-----+-----+-----+-----+
| student_name | Roll_number | mark1 | mark2 | mark3 | mark4 | total |
+-----+-----+-----+-----+-----+-----+
| Rajesh      | 101         | 70    | 64    | 53    | 90    | 277    |
+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

mysql> select * from student_mark where NOT MARK1<=90;
Empty set (0.01 sec)

mysql> select * from student_mark where NOT MARK1=90;

```
+-----+-----+-----+-----+-----+-----+
| student_name | Roll_number | mark1 | mark2 | mark3 | mark4 | total |
+-----+-----+-----+-----+-----+-----+
| Rajesh      | 101         | 70    | 64    | 53    | 90    | 277    |
| Isai        | 104         | 56    | 45    | 74    | 80    | 255    |
| Saran       | 105         | 24    | 67    | 43    | 60    | 194    |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

mysql>

SEARCH OF DATA BY DIFFERENT CRITERIA

SEARCH OF DATA BY DIFFERENT CRITERIA

SORTING OF DATA

```
mysql> SELECT * FROM student_mark ORDER BY Roll_number desc;
```

student_name	Roll_number	mark1	mark2	mark3	mark4	total
Saran	105	24	67	43	60	194
Isai	104	56	45	74	80	255
jagan	103	90	75	96	99	360
Rajesh	101	70	64	53	90	277

```
4 rows in set (0.02 sec)
```

```
mysql> SELECT * FROM student_mark ORDER BY Roll_number asc;
```

student_name	Roll_number	mark1	mark2	mark3	mark4	total
Rajesh	101	70	64	53	90	277
jagan	103	90	75	96	99	360
Isai	104	56	45	74	80	255
Saran	105	24	67	43	60	194

```
4 rows in set (0.00 sec)
```

UPDATE TABLE

```
mysql> Alter table student_mark add(total int(20));
```

Query OK, 0 rows affected (1.09 sec)

Records: 0 Duplicates: 0 Warnings: 0

```
mysql> update student_mark set total=mark1+mark2+mark3+mark4;
```

Query OK, 5 rows affected (0.22 sec)

Rows matched: 5 Changed: 5 Warnings: 0

```
mysql> select * from student_mark;
```

student_name	Roll_number	mark1	mark2	mark3	mark4	total
Rajesh	101	70	64	53	90	277
Karthiga	102	90	85	84	100	359
jagan	103	90	75	96	99	360
Isai	104	56	45	74	80	255
Saran	105	24	67	43	60	194

5 rows in set (0.00 sec)

```
mysql> _
```